Pachydermoperiostosis Mimicking Acromegaly: A Case Report

Abstract

Pachydermoperiostosis is a rare, hereditary disease commonly presenting with digital clubbing, pachyderma, and periosteal hypertrophy. Coarsening of facial features and spade-like enlargement of hands and feet may give rise to a diagnostic dilemma between pachydermoperiostosis and acromegaly. This report highlights a case of a 36-year-old man who presented with broadening of hands and feet, facial skin thickening, and edematous and drooping eyelids for the last 10 years. There was no history of similar presentation in his family. Such clinical presentation in corroboration with normal growth hormone level and prominent radiological abnormalities prompted us to make a diagnosis of pachydermoperiostosis.

Keywords: *Cutis verticis gyrate, hypertrophic osteoarthropathy, pachyderma, pachydermoperiostosis, Touraine–Solente–Gole syndrome*

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Introduction

Pachydermoperiostosis, also known as Touraine–Solente–Gole syndrome, is the primary idiopathic form of hypertrophic osteoarthropathy (HOA) characterized by clubbing of the digits, enlargement of the extremities, and painful joints. It is a rare genodermatosis with a prevalence of 0.16%.[1] Thickening of the skin on the face and scalp with coarsening of facial features, seborrhea, and hyperhydrosis may be present. The disorder is inherited as an autosomal dominant trait and is seven times more common among boys than girls. The folding of the scalp skin results in cutis verticis gyrata. Thickening of the phalanges and of the bones of the limbs produces spade-like hands and feet and hypertrophic osteoarthropathy. In pachydermoperiostosis, no causative factors such as pulmonary, cardiac, or hepatic disorders have been demonstrated.[2] Three forms have been described –a complete form with pachyderma and periostitis, an incomplete form with evidence of bone abnormalities but lacking pachyderma, and a *forme frusta* with prominent pachyderma and minimal-to-absent skeletal changes.

Case Report

A 36-year-old Muslim nonalcoholic male patient borne out of a consanguineous marriage presented to our OPD with a 10-year history of progressive thickening of facial skin with longitudinal and transverse folds resembling leonine facies. He also gave a history of visual disturbances because of drooping and thickening of eyelids. There was history of intermittent joint pain in wrists, elbows, and ankles. He had no other pertinent medical or familial history.

On examination, the patient had pronounced folds in the area of forehead, between the eyes, in the nasolabial grooves, and on the chin [Figure 1]. His nose was excessively enlarged with thickened skin folds [Figure 2]. The skin was greasy with excessive seborrhoea and sweating. The skin of the hand and foot was also thickened with severe clubbing of all the fingers and toes [Figures 3 and 4]. Cutis verticis gyrata was not present. Both ankles were mildly swollen and his blood pressure was normal.

Laboratory analyses showed normal erythrocyte sedimentation rate, C-reactive protein, random blood sugar and serum calcium, insulin-like growth factor-1 (IGF-1), thyroid function tests, and pepsinogen-1 with a nonreactive VDRL test. His growth hormones were normally suppressed with glucose (75 g) loading. Negative results were obtained for rheumatoid factor, anticyclic citrullinated

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Figure 1:Increase in skin folds, enlarged, and thickened nose



Figure 3:Clubbing of great toe

protein (anti-CCP) antibodies, and HLA-B27. Slit skin smear and skin biopsy with Fite acid stain were done, and no AFB were found on both occasions. X-rays revealed periosteal thickening of the long bones in the diaphyses of metatarsal and metacarpals as well as along cortices of long bones [Figures 5 and 6]. Chest X-ray shows enlarged clavicles, cardiomegaly with normal lung markings. Radiograph of the skull and CT scan of pituitary fossa were normal.



Figure 2:Coarse facial features



Figure 4:Clubbing of fingers

Discussion

Pachydermoperiostosis is a rare hereditary disorder. The diagnostic criteria for pachydermoperiostosisinclude:

- Major criteria: pachyderma, periostosis, finger clubbing.
- Minor criteria: hyperhidrosis, arthralgia, gastric ulcer, cutis verticis gyrata, blepharoptosis, joint effusion, column-like legs, edema, seborrhea, acne, flushing.[3]

Our patient had all three major criteria, i.e., hyperostosis, finger clubbing, and pachyderma. Usually, in such cases, symptoms beginin teenage years but in our case the symptoms began in the late twenties.[4] Pachydermoperiostosis is the primary form of HOA; Secondary HOAs exist with underlying diseases such as malignancy.[5] Patients may also present with seborrhea (90% of cases), acne, folliculitis, dilated pores, hyperhidrosis of the palms, and soles (44–67% of cases), flushing, thickened eyelids (30–40% of cases), and cutis vertices gyrate (24% of cases). In our case, there were seborrhea, hyperhidrosis, thickened, and drooping eyelids.



Figure 5:Periosteal thickening of diaphyses of metacarpals



Figure 6:Periosteal thickening of metatarsals and cortices of long bones

Martinez-Lavin proposed that the pathology for pachydermoperiostosis is due to increased amount of proliferation of collagen fibers from actively proliferating fibroblasts.[6] Wegrowski proposed the pathology for pachydermoperiostosis as dysregulation of the matrix deposits and increased synthesis of decorin protein.[7] Some studies explained the role of epidermal growth factor, Von Willebrand factor antigen, vascular endothelial growth factor and platelet derived growth factor in the pathogenesis of pachydermoperiostosis.[8] Mutations in the *HPGD* gene on chromosome 4q34.1 have been found in those with the autosomal recessive form of this condition.[9]

The clinical and radiological presentations in pachydermoperiostosis can be confused with those of syphilitic periostitis, psoriatic onychopachydermoperiostitis (POPP), Paget's disease, and especially, acromegaly.[10] The clues for the differential diagnosis of pachydermoperiostosis are digital clubbing and periostosis, which are not features of acromegaly. Symptomatic management of arthralgia is done by nonsteroidal anti-inflammatory drugs, systemic or intraarticular steroids, and oral tamoxifen. Colchicine improvethe articular symptoms, folliculitis, and pachyderma. Rheumatologic symptoms have been improved by intravenous pamidronate. Isotretinoin showed improvement of the skin symptoms with pachyderma and cutis verticis gyrate.

Conclusion

Pachydermoperiostosis cases mimicking acromegaly have been reported to bring them to the attention of practitioners. Although both acromegaly and pachydermoperiostosis are infrequently encountered, avoidance of diagnostic confusion is important because of the prognostic and therapeutic implications. Awareness of the significance of clubbing under these circumstances is likely to prevent misdiagnosis.

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Conflicts of interest

There are no conflicts of interest.

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